ModelIt!

Teacher's Guide

Why Are There So Many Kinds of Bugs?

Exploring the Diversity of Life on Earth

Quick Reference

Grade Level	Grade 2
NGSS Standard	2-LS4-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 2-LS4-1

Make observations of plants and animals to compare the diversity of life in different habitats.

Three-Dimensional Learning

Science Practice	**Making Observations** — Students observe and compare insects in different habitats, noting differences in body features and behavior.
Disciplinary Core Idea	**LS4.D Biodiversity and Humans** — There are many different kinds of living things in any area, and they exist in different places. The diversity of life is connected to the diversity of habitats.
Crosscutting Concept	**Structure and Function** — Students connect insect body features (structure) to how those features help the insect survive (function) in a specific habitat.

CAST Testing Alignment

MULTIPLE CHOICE:

A water beetle has flat paddle-shaped legs. A grasshopper has long strong jumping legs. Why are their legs so different?

CONSTRUCTED RESPONSE:

Choose two insects from different habitats. Explain how their body features help each one survive in its own habitat.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Insects are the most diverse group of animals on Earth. There are more species of insects than all other animals combined! Understanding why insects are so diverse teaches us about how life adapts to different environments.

The World of Insects

Scientists have discovered over one million species of insects, and they estimate there may be millions more waiting to be found! Insects live on every continent including Antarctica. They are found in gardens, forests, deserts, ponds, caves, and even inside our homes. Insects are incredibly diverse because they have adapted to nearly every habitat on Earth.

Body Features Match Habitats

Each insect species has body features that help it survive in its specific habitat. Butterflies have long tongues for reaching nectar deep inside flowers. Water beetles have flat, paddle-shaped legs for swimming. Stick insects are shaped and colored exactly like twigs to hide from predators. Walking sticks, leaf insects, and bark beetles all use camouflage to blend into their surroundings. Each feature is an adaptation — a body part that helps the animal survive.

Why So Many Species?

There are so many insect species because there are so many different habitats and food sources on Earth. Each unique habitat creates opportunities for insects with different features. A pond needs swimmers. A garden needs fliers. A forest floor needs diggers. Over millions of years, insects have evolved different features for different habitats, creating the incredible diversity we see today.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify habitat type as an external component and body features and survival success as internal components that are determined by the habitat's challenges.

E **Establish Relationships**
Students determine that different habitat types produce different body features (the habitat shapes which features are useful), and better-matched features lead to higher survival success.

V **Visualize & Model**
Students build a model showing how the type of habitat determines which insect body features lead to the best survival.

E **Evaluate Outcomes**
Students run garden, pond, and forest floor scenarios to compare how different habitats produce different insect features.

R **Revise & Extend**
Students add predator type or food source to explore more reasons why insects in different habitats look different.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"How many different kinds of bugs can you think of? Beetles, butterflies, ants, dragonflies — why are there SO many?"*

Watch For

Show a poster or slideshow of 8-10 very different-looking insects. Ask: Why do they all look so different?

Slide 2: What Will We Learn Today?

3 min

Say: *"Today we are going to figure out why there are so many different bugs and why they all look different!"*

Watch For

Read objectives together. Introduce diversity, habitat, insect, and feature with picture cards.

Slide 3: The Big Question

3 min

Say: *"There are more than a MILLION different kinds of insects! Why so many? And why do they all have different bodies?"*

Watch For

Show a butterfly and a water beetle side by side. Ask: Why does one have wings and the other has paddles?

Slide 4: Let's Build a Model!

3 min

Say: *"We are going to build a model that shows how habitats create different kinds of insects."*

Watch For

Preview LEVER steps. Show pictures of a garden, a pond, and a forest floor. Ask: Would the same bug survive in all three?

Slide 5: Activity 1: Sort the Parts

7 min

Say: *"The habitat is where the insect lives. Its body features help it survive there. Let us sort our components!"*

Watch For

Guide sorting: Habitat Type is external (it already exists), Body Features and Survival Success are internal.

Slide 6: Activity 2: Connect with Arrows

7 min

Say: *"When we change from a garden to a pond, do the insect features stay the same or change? Why?"*

Watch For

Draw arrows from Habitat Type to Body Features (habitat determines needed features). Draw arrow from Body Features to Survival Success.

Slide 7: Activity 3: Run the Simulation!

8 min

Say: *"Let us test! What features does an insect need in a garden? A pond? A dark forest floor?"*

Watch For

Students predict which features match each habitat, then run simulations. Compare insects across habitats.

Slide 8: What Did We Discover?

7 min

Say: *"Now we know! There are so many kinds of bugs because there are so many different places to live. Each habitat needs different features!"*

Watch For

Show the million-species statistic. Have students share which habitat-insect match surprised them most.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Habitat Type
- **INTERNAL:** Body Features, Survival Success

Relationships

- **Habitat Type to Body Features:** POSITIVE (+) — Different habitats produce different body feature needs. A pond habitat needs swimming features. A garden habitat needs flying and camouflage features. The habitat shapes which features are most useful for survival.
- **Body Features to Survival Success:** POSITIVE (+) — When an insect has body features that match its habitat well, survival success is high. A water beetle with paddle legs thrives in a pond. A butterfly with wings thrives in a garden. Good feature-habitat match means better survival.

Key Discoveries

- Different habitats have different kinds of insects because each habitat requires different survival features
- Garden insects often have wings, bright colors for warning, or camouflage to blend in with plants
- Pond insects have features for swimming or walking on water, like flat legs or air bubbles
- The diversity of insects exists because there are so many different habitats, each needing different adaptations

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"All bugs are the same"	Insects are incredibly diverse — there are over a million different species! A ladybug, a dragonfly, and an ant may all be insects, but they have very different bodies, eat different foods, and live in different places. Their differences help them survive in their own specific habitats.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Bugs live everywhere and do not need a specific home"	Every insect species has a habitat it is best suited for. A water beetle cannot survive in a desert, and a desert beetle would struggle in a pond. Insects need specific conditions — temperature, food, water, shelter — that their body features are designed for.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Insects choose to look the way they do"	Insects do not choose their body features — they are born with them! Over many, many generations, insects that had features matching their habitat survived better and had more babies. Those babies inherited the same helpful features. This is how species develop features that match their habitats over a very long time.